An In-Silico Pipeline for CRISPR–Cas9 Guide Design Against the HIV-1 Proviral Genome: Cross-Subtype Conservation, Genome-Wide Human Off-Target Exclusion, and Transcriptome Similarity Screening

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Abstract

Antiretroviral therapy suppresses but does not eradicate HIV-1, and the first-in-human CRISPR excision trial (EBT-101) was safe but did not provide durable suppression after treatment interruption. Target and guide-RNA selection therefore remain open design problems. We built a fully scripted, reproducible pipeline that (i) star-aligns 24 diverse HIV-1 complete genomes to the HXB2 reference to quantify cross-strain conservation of every candidate guide; (ii) exhaustively screens guides against GRCh38 for off-target cleavage sites with up to four mismatches (both strands; NGG/NAG for SpCas9, NNGRRT/NNGRRV for SaCas9); and (iii) measures exact k-mer sharing between the HIV-1 genome and both the human genome and the human protein-coding transcriptome. Of 1,365 enumerated protospacers, 668 passed design filters and 190 (150 SpCas9, 40 SaCas9) underwent genome-wide screening. No guide had a single exact-match off-target in the 3.1-billion-base human genome; 63 guides carried no off-target with $\leq$2 mismatches and no transcriptome hit with $\leq$1 mismatch (tier A+). Region-level analysis identifies the 5′ LTR U5 region and the packaging-signal leader as the combination of high conservation and minimal human similarity, whereas *pol* and *env* carry substantially more human-shared 20-mers, consistent with endogenous-retrovirus ancestry. Notably, guides in the otherwise most conserved leader region exactly match human *NOTCH3* transcripts and were excluded. We recommend a dual-guide design (5′LTR U5 + integrase; 4,439 bp excision) with a triple-guide escape-hardened variant, and provide all sequences, off-target tables, and the evidence ledger.

1 Introduction

Combination antiretroviral therapy (ART) controls plasma viremia but cannot reach the latent proviral reservoir, which re-seeds viremia when treatment stops. Programmable nucleases offered a radical alternative: rather than blocking replication, excise the proviral DNA itself. Early studies showed that CRISPR–Cas9 can disrupt or excise integrated HIV-1 DNA in cell lines[1], primary T cells[6], and across the LTR–gag region[2], and that multiplexed SaCas9 guide RNAs delivered by

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AAV can excise proviral fragments in animal models[9]. Combining long-acting ART with CRISPR excision eliminated infectious virus in a subset of humanized mice[4].

This line of work has reached humans. EBT-101 (Excision BioTherapeutics) is an AAV9 vector carrying SaCas9 with two guide RNAs targeting the LTR and gag (construct "AAV9-SaCas9-GagD-LTR1")[55,20]. Its phase 1/2 trial (NCT05144386) enrolled six participants, completed in November 2024[50,55], and reported no serious adverse events and no identified off-target DNA damage[18,29]. However, HIV RNA rebounded in all participants who interrupted ART[29,20,21], and the program is listed as inactive with the curator note "safe, but does not provide long-term viral suppression"[55]. The lesson is not that proviral excision failed as a concept, but that target selection, guide quality, and reservoir coverage were insufficient[22,47]. Two design questions follow directly: which viral regions should be cut, and how much do candidate guides resemble the human genome they must not cut?

This paper addresses both questions computationally. We ask: *which part of the HIV-1 genome is the best CRISPR target, considering cross-strain conservation, functional importance documented in the literature, and least similarity to human genes?* We contribute: (1) a reproducible pipeline that quantifies conservation across 24 diverse complete genomes; (2) an exhaustive human-genome off-target scan (to four mismatches, both strands, canonical and non-canonical PAMs) implemented in C and validated guide by guide; (3) a transcriptome similarity screen that catches human-exact guide sequences that whole-genome DNA screens under-weight; and (4) a ranked target-region table and concrete dual/triple guide designs.

2 Background and related work

2.1 Target regions used by published studies

Published anti-HIV-1 CRISPR work has targeted the long terminal repeats (LTRs)[1,6,2], gag[9], and combinations thereof. Because the integrated provirus is flanked by two LTRs that are identical copies of the same sequence, a single LTR-targeting guide cleaves both ends, and NHEJ re-joining excises the intervening genome[9]. Yin et al. excised proviral DNA in vivo with SaCas9 and multiplex gRNAs[9], and Dash et al. estimated excision of approximately 80% of 5′LTR–gag and gag–3′LTR fragments in humanized mice, reporting "No CRISPR-Cas9 mediated off-target effects are detected"[4].

2.2 Viral escape

Cas9 pressure selects for resistant virus: NHEJ repair of the cut site inserts indels, and escape variants emerge rapidly when non-conserved regions are attacked (breakthrough replication in 11–17 days) and more slowly for conserved targets (20–43 days)[32,3]. Resistance mutations cluster at the cleavage site[3], and host double-strand-break repair directly generates CRISPR-resistant strains[28]. The standard mitigation is multiplexing several conserved, independent targets[3,22,47].

2.3 Off-target risk

Subtype-specific off-target analyses of LTR guide collections found that 149 of 4,114 viable guides (3.60%) retained predicted human off-target cleavage, of which only 14 of 326 sites (4.29%) were exonic[11]. Unbiased assays (GUIDE-seq, CIRCLE-seq, whole-genome sequencing) have repeatedly failed to detect notable off-target cleavage for anti-HIV-1 guides[27], and clinical monitoring of EBT-101 participants likewise identified no off-target DNA damage[29]. Human endogenous retroviruses

(HERVs) share ancestry with *gag* and *pol*, but an analysis of predicted off-targets concluded HERV loci are "not the driving factor for off-target editing" for currently used guides[27]; our region-level similarity data (Section 4.3) add quantitative context.

2.4 RNA-targeting alternatives

Cas13 systems target viral RNA without cutting DNA: LbuCas13a reduced HIV-1 RNA by 50–80%[8], and RfxCas13d with crRNAs against conserved *gag/pol* regions inhibited replication by more than 90% in primary $CD4^+$ T cells[44,16]. Combined RNA- and DNA-targeting ribonucleoproteins are an active frontier[54]. Our pipeline designs Cas9 guides; Cas13 candidates are out of scope but the same conservation/off-target machinery applies.

3 Methods

3.1 Reference, regions, and strain set

All coordinates use the HXB2 reference genome (GenBank K03455, 9,719 bp) and its GenBank feature annotation[48,49], cross-checked against the LANL annotated HXB2 map with per-position Shannon entropy[51]. We define 31 regions: LTR subregions (U3/R/U5, both copies), TAR, the packaging signal ψ, *gag* (p17, p24), *pol* (protease, RT, RNase H, integrase, cPPT), *vif*, *vpr*, *tat*, *rev*, *vpu*, *env* (gp120, RRE, gp41), and *nef*. The ψ coordinates remain literature-derived and unverified; all others come from the GenBank record or the LANL map. Note that LANL and GenBank label R-region boundaries within ±2 nucleotides of each other; we use GenBank throughout.

We downloaded 24 near-full-length complete HIV-1 genomes (8.5–9.9 kb) from NCBI via E-utilities, selected for country diversity across Africa, Europe, Asia, and the Americas, with subtype labels taken verbatim from record titles where present, and excluding records titled as mutants, synthetic constructs, or vectors. The NL4-3 proviral genome was extracted from the pNL4-3 molecular clone record[58]. Together with HXB2 this yields 25 genomes for conservation scoring.

3.2 Conservation

Each strain was globally aligned to HXB2 (Biopython PairwiseAligner; match +2, mismatch −1, gap open −4, extension −1), producing a base at every HXB2 position (gaps/ambiguity coded separately). For each candidate guide, conservation is the fraction of strains whose aligned sequence matches the HXB2-derived guide across the full protospacer *and* whose PAM matches at the constrained PAM positions — i.e., the fraction of strains in which the guide site is present unmutated.

3.3 Guide enumeration and filters

Protospacers follow the standard convention: the ordered guide sequence is the 20-nt (SpCas9) or 21-nt (SaCas9) sequence immediately 5′ of the PAM on the PAM-bearing strand. We enumerated every SpCas9 NGG site and every canonical SaCas9 NNGRRT site on both strands (1,365 sites). Design filters: GC content 40–65%, no homopolymer run of four or more identical bases (this also enforces the U6 promoter poly-T termination rule of standard guide design[38,35]), and no TTT start. Surviving guides (544 SpCas9, 124 SaCas9) were ranked by conservation, then GC near 50%; the top 150 SpCas9 and top 40 SaCas9 advanced to genome-wide screening.

3.4 Human off-target scan

A purpose-built C tool (`crispr_scan`; pthreads, 4-bit packed genome) exhaustively scanned GRCh38 (Ensembl primary assembly, release 113[56]) for all sites with Hamming distance ≤ 4 to any guide with a valid PAM: NGG and NAG for SpCas9, NNGRRT and NNGRRV for SaCas9, both strands, with N bases always counted as mismatches. The tool passes 13 unit tests including hand-derived strand/PAM vectors, and every screened guide was confirmed to hit HXB2 exactly at its expected coordinate and strand (190/190), validating the conventions end-to-end. The full 3.1-Gbp scan of 190 guides completes in approximately five minutes on six ARM cores.

Safety tiers. A random 20-mer has roughly 30 by-chance $\leq$3-mismatch sites in a 3.1-Gbp genome, so "zero $\leq$3 mm sites" is not a meaningful bar. We therefore tier guides by the sites that matter: **A+** — no genomic off-target with $\leq$2 mismatches *and* no transcriptome hit with $\leq$1 mismatch; **A** — no genomic off-target with $\leq$1 mismatch and no exact transcriptome hit; **B** — everything else. Cleavage at $\geq$3 mismatches is weak, and sites at 1–2 mismatches are the residual risk band[34,27].

3.5 Human-similarity maps

We measured exact k-mer sharing between the HXB2 genome and (a) GRCh38 ($k = 20$ and $k = 30$) and (b) GENCODE v44 protein-coding transcripts[57] ($k = 20$), using the same packed-scanning tool (rolling k-mer, both orientations, N-excluded). For every off-target site with $\leq$2 mismatches (490 sites), we additionally scanned the protein-coding transcriptome at $\leq$1 mismatch to find guide sequences that could cleave human mRNAs.

4 Results

4.1 Conservation across 25 strains

Per-position identity to HXB2 ranged from 80.2% to 97.3% (NL4-3, the closest subtype-B genome). Table 1 summarizes each region's median guide conservation alongside the independent LANL entropy values[51]: the two measures agree (integrase and cPPT are the most conserved coding features in both). The 5′ LTR U5 region and the R region score highest among designable regions (median guide conservation 0.72 and 0.84), and the leader/ψ region reaches 0.56 with individual guides at 0.96.

4.2 Genome-wide off-target screen

Across all 190 screened guides there were **zero exact-match (mm=0) off-target sites** in GRCh38. The full mismatch distribution was: 20 sites at 1 mismatch (17 guides), 470 at 2, 6,773 at 3, and 76,105 at 4. Tiers: 63 A+, 102 A, 25 B. The single-mismatch sites are listed per guide in the repository; no guide carries more than two.

4.3 Human-similarity maps

The HIV-1 genome is strikingly alien to human DNA: **not a single exact 30-mer is shared** between HXB2 and the 3.1-Gbp human genome, and only 261 of 9,700 possible 20-mer start positions (2.7%) share an exact 20-mer with it. Sharing is concentrated where retroviral ancestry predicts: RT (53 human-shared 20-mer positions), gp120 (42), integrase (40), gp41 (31), and p17 (14), versus 1 for 5′LTR U5, 0 for ψ and 3′LTR U5 (Table 1).

Table 1: HXB2 regions (GenBank[48] / LANL map[51] coordinates), median guide conservation across 25 strains, LANL per-region mean Shannon entropy[51] where published, number of tier-A+ guides, and exact 20-mer sharing with the human genome (hu) and protein-coding transcriptome (tx). †literature-derived, unverified.

Region	HXB2 span	Source	Med. cons.	LANL ent.	A+ guides	hu 20-mer	tx 20-mer
TAR	454–513	LANL map	0.16	–	0	2	0
R_5	454–551	genbank	0.84	–	1	2	0
U3_5	1–453	genbank	0.00	–	2	9	0
U5_5	552–634	genbank	0.72	–	5	1	1
psi	744–802	lit.†	0.92	–	0	0	0
p17	791–1186	LANL map	0.20	0.3664	8	14	0
p24	1186–1878	LANL map	0.20	0.2073	12	7	0
protease	2253–2549	LANL map	0.20	0.2002	0	5	0
RT	2550–3869	LANL map	0.12	0.2022	6	53	2
RNaseH	3870–4229	LANL map	0.12	0.2214	0	17	0
IN	4230–5093	LANL map	0.12	0.1598	4	40	2
cPPT	4781–4785	LANL map	–	0.0112	0		
vif	5041–5619	genbank	0.16	0.2600	4	3	0
vpr	5559–5795	genbank	0.04	0.2616	1	1	0
tat_e1	5831–6045	genbank	0.24	0.3129	2	2	0
rev_e1	5970–6045	genbank	–	0.3299	0		
vpu	6062–6310	LANL map	0.04	0.4378	0	12	2
gp120	6309–7757	LANL map	0.08	0.4433	2	42	0
RRE	7710–8061	LANL map	0.12	0.2156	6	8	0
gp41	7758–8792	LANL map	0.04	0.3234	3	31	1
nef	8797–9168	genbank	0.04	–	0	7	0
U3_3	9086–9539	genbank	0.04	–	2	6	0
R_3	9540–9636	genbank	0.24	–	1	2	0
U5_3	9637–9719	genbank	0.44	–	3	0	0

The transcriptome screen proves this matters at guide resolution: among the 490 off-target windows with ≤2 mismatches, 225 have a ≤1-mismatch match in protein-coding transcripts. Critically, the two most conserved guides in the entire genome — SP_772+ and SP_771+ in the packaging-signal leader, 0.96 and 0.92 conserved across strains — carry *exact* (mm=0) matches to human *NOTCH3* transcripts (NOTCH3-201) and would have passed a conservation-first design. They were excluded by the transcriptome screen. Other exact transcript matches involve *NFX1*, *CHRNA2*, and olfactory-receptor loci.

4.4 Recommended designs

Table 2 lists the recommended multiplex designs, all tier A+. The dual design pairs SP_628+ (5′LTR U5, 0.88 conserved, zero off-targets ≤2 mm) with SA_5067+ (integrase, 0.48) for a 4,439 bp excision; because in vivo proviral LTRs are identical, the U5 guide cuts both LTR ends, and the triple design adds SP_7794+ (RRE, 0.76) so that three independent conserved sites must mutate simultaneously for escape, directly implementing the multiplexing mitigation[3,32,22]. Runners-up for the internal slot were p24 guides (12 tier-A+ guides, 7 human-shared 20-mers, zero transcriptome hits), vif (0.44), and p17 (0.44).

Table 2: Recommended tier-A+ guide combinations (protospacer/PAM as read on the PAM-bearing strand). Cons. = fraction of 25 strains with the guide site unmutated.

Strategy	Guides (protospacer/PAM)	Excised span	Min. cons.
dual excision	TCTAGCAGTGGCGCCCGAAC–AGG (SP_628+) TGTGTGGCAAGTAGACAGGAT–GAGGAT (SA_5067+)	4,439 bp	0.48
triple-guide escape-hardened excision	TCTAGCAGTGGCGCCCGAAC–AGG (SP_628+) TGTGTGGCAAGTAGACAGGAT–GAGGAT (SA_5067+) GGAGCAGCAGGAAGCACTAT–GGG (SP_7794+)	multi-fragment (three cut sites)	0.48

Table 3: Top tier-A+ guides by conservation. Full tables for all 190 guides, including every off-target site and its coordinate, are in the repository.

ID	Region	Position	Protospacer ($5' \to 3'$)	PAM	System	Cons.	mm≤4
SP_628+	U5_5	628 +	TCTAGCAGTGGCGCCCGAAC	AGG	SpCas9	0.88	72
SP_629+	U5_5	629 +	CTAGCAGTGGCGCCCGAACA	GGG	SpCas9	0.88	73
SP_7794+	RRE	7794 +	GGAGCAGCAGGAAGCACTAT	GGG	SpCas9	0.76	442
SA_584-	U5_5	584 -	TCCACAGATCAAGGATATCTT	AGATGG	SaCas9	0.72	13
SP_8568+	gp41	8568 +	GTGGAACTTCTGGGACGCAG	GGG	SpCas9	0.68	316
SP_8567+	gp41	8567 +	TGTGGAACTTCTGGGACGCA	GGG	SpCas9	0.68	337
SP_796+	p17	796 +	GCGAGAGCGTCAGTATTAAG	CGG	SpCas9	0.56	63
SP_562+	U5_5	562 +	GCCCGTCTGTTGTGTGACTC	TGG	SpCas9	0.52	143
SP_566-	U5_5	566 -	AGCCTTGTGTGTGGTAGATC	TGA	SpCas9	0.52	397
SA_5067+	IN	5067 +	TGTGTGGCAAGTAGACAGGAT	GAGGAT	SaCas9	0.48	79
SA_4382+	IN	4382 +	TGGACAAGTAGACTGTAGTCC	AGGAAT	SaCas9	0.44	87
SP_9647+	U5_3	9647 +	GCCCGTCTGTTGTGTGACTC	TGG	SpCas9	0.44	143
SP_5199+	vif	5199 +	AGAAGTACACATCCCACTAG	GGG	SpCas9	0.44	204
SP_884+	p17	884 +	AACATATAGTATGGGCAAGC	AGG	SpCas9	0.44	220

5 Discussion

The sweet spot is the LTR U5 region, with p24 as the best internal partner. Our data resolve the design tension cleanly. The most *conserved* guides sit in the leader/ψ region (0.96) — but the most conserved guides are exactly the ones that match human transcripts (*NOTCH3*), a trap for conservation-first pipelines. The $5'$ LTR U5 region offers near-top conservation (0.72–0.88) with essentially no human similarity (one human-shared 20-mer position genome-wide), the dual-cut excision geometry of the LTRs, and five tier-A+ guides. Among internal regions, p24 combines the largest tier-A+ pool (12) with low human sharing, whereas *pol* and *env* — despite excellent functional essentiality — carry the most human-shared sequence, as expected from HERV ancestry; this did not produce off-targets for our specific guides (consistent with[27]), but it lowers the margin of safety for guides designed naively in those regions.

Escape remains the binding constraint. EBT-101's rebound after treatment interruption[29,20,21] and the documented speed of NHEJ-driven escape[32,3] both point to the same

conclusion: two guides are likely insufficient, and reservoir coverage — not guide safety — is the hard problem. Proviral DNA circles with reformed LTRs can persist after excision[13], defective proviruses accumulate from acute infection onward[15], and intact inducible proviruses are the true target population[39]. Our triple design (LTR U5 + integrase + RRE) maximizes simultaneous-escape requirements while keeping every guide in tier A+; whether three guides suffice against the reservoir is an empirical question that only animal and clinical work can answer[22,47,43].

Relationship to EBT-101. EBT-101 targets GagD and LTR1 with SaCas9[55,20]. Our pipeline's top LTR guide differs (U5 rather than their LTR site) because we required zero $\leq$2 mm off-targets and transcriptome cleanliness; whether the published GagD/LTR1 guides would pass our tier system is directly testable with the repository pipeline and is planned as a benchmark. The clinical experience to date — safe, no off-target damage, but viral rebound[18,29,50] — is consistent with our modeling that guide safety is achievable and that reservoir coverage and escape are the unsolved problems.

6 Limitations

This is a computational design study; nothing here is experimentally validated, and nothing here is medical advice. Specific limitations: (1) off-target scanning covers mismatches only, not DNA or RNA bulges, and uses NGG/NAG and NNGRRT/NNGRRV PAM sets — CHOPCHOP/CRISPOR-grade re-scoring with bulge models is advisable before any wet-lab use; (2) conservation uses 24 near-full-length genomes selected for geographic spread, not a curated subtype-balanced panel, and subtype labels come from record titles (only two records carried explicit subtype labels); (3) the transcriptome screen covers protein-coding transcripts only (GENCODE v44); non-coding RNAs are unscreened; (4) ψ coordinates are literature-derived and unverified; LANL and GenBank differ by $\pm$2 nt on some LTR feature boundaries; (5) SaCas9's NNGRRT PAM yields few designable sites (124 passing), limiting AAV-compatible options relative to SpCas9; (6) no chromatin, nucleosome, or delivery context is modeled; and (7) viral escape dynamics, reservoir size, and excision efficiency are not modeled. In silico tier A+ status is a necessary, not sufficient, condition for a therapeutic guide.

7 Data and code availability

The repository (`crispr-hiv-ward`) contains the full pipeline (`src/`), unit tests (39 passing across the Python pipeline and the C scanner), all result tables (`results/`), the strain sequences, packed-genome tooling, the citation ledger with verbatim source quotes (`docs/citations/`), and this paper's LaTeX source. The human genome (Ensembl release 113[56]) and GENCODE v44[57] are fetched by script; strain sets are fetched from NCBI E-utilities with reproducible queries.

Acknowledgments

Computational work was performed on a NVIDIA Jetson Orin Nano (ARM64, 8 GB) running fully scripted, reproducible tooling; all code, data, results tables, and the citation ledger are in the accompanying repository (`crispr-hiv-ward`). Hermes Agent is an autonomous research agent built by **Nous Research**; it carried out the pipeline engineering, source retrieval and verification, analyses, and manuscript drafting under Dr. Kirkpatrick's direction, with all findings and design criteria

subject to his review and approval. We thank the maintainers of NCBI, Ensembl, GENCODE, and the Los Alamos HIV Database for open access to the sequence data and annotations that made this work possible.

Sources

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Quote: "No CRISPR-Cas9 mediated off-target effects are detected."
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Quote: "NAs with predicted off-target cleavage was calculated to be 149 of the 4,114 viable gRNAs that were created, which corresponds to 3.60% of the tested gRNAs."
Quote: "Of the 326 off-target effects, only 14 (4.29%) of them were exonic"
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Quote: "Completion Date | 2024-11-14"

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57 GENCODE release 44: protein-coding transcripts (data source)
https://ftp.ebi.ac.uk/pub/databases/gencode/Gencode_human/release_44

58 GenBank M19921: pNL4-3 molecular clone record (NL4-3 proviral genome extracted)
https://www.ncbi.nlm.nih.gov/nuccore/M19921